

Study Guide, Exercises, and Sample Questions

CSE 4615: Wireless Networks

Islamic University of Technology (IUT)

Lecture 5: IEEE 802.11 Wireless Local Area Networks

1. Topics Covered

- Scope and overview of the IEEE 802.11 standard.
- 802.11 Architecture: Basic Service Set (BSS), Independent Basic Service Set (IBSS), Distribution System (DS).
- Coordination Function (CF): role in managing access to the wireless medium.
- Distributed Coordination Function (DCF): distributed medium access for asynchronous traffic; no station priority.
- Point Coordination Function (PCF): optional extension for QoS support (synchronous traffic).
- All 802.11 variants use CSMA/CA for multiple access.
- DCF operation: CSMA-based listen-before-talk with no collision detection; use of CSMA/CA and ARQ (link-layer ACK/retransmission).
- Clear Channel Assessment (CCA): physical carrier sensing using a power threshold (-82 dBm); variants including ordinary noise detection and 802.11-only signal detection.
- Network Allocation Vector (NAV): virtual carrier sensing; channel reservation timer that decrements independently of medium status.
- Timing and Interframe Spaces: aSlotTime, and the four IFS types – SIFS, PIFS, DIFS, and EIFS – their values in 802.11a and their priority hierarchy.
- Frame exchange scenarios: first unsuccessful MSDU delivery, second unsuccessful MSDU delivery (ACK lost), and successful MSDU delivery.
- Collision avoidance mechanism: the backoff procedure, Contention Window (CW), and the formula $CW = \min(2^{(n-1)} \cdot CW_{\min} - 1, CW_{\max})$.
- Binary Exponential Backoff (BEB): doubling of CW after each failed transmission attempt; frozen countdown during busy channel; resumption of countdown after idle channel.
- RTS/CTS exchange in the context of 802.11: channel reservation, NAV setting by neighbors, and protection against hidden stations.
- 802.11 sender and receiver state machines: complete DCF procedure including back-off, RTS, CTS, DATA, and ACK.

- 802.11 Medium Access Overhead: DIFS, random backoff, RTS/CTS, ACK/SIFS, and header overhead; channel wastage due to backoff and collisions; approximately 35% overhead at 54 Mbps.
- Significant inefficiencies of 802.11 DCF: idle channel time during countdown, collision probability with many nodes, and exponential backoff under congestion.
- Hidden stations and RTS/CTS: NAV set upon receiving RTS (by neighbors of transmitter) or CTS (by hidden stations near receiver); role of SIFS in prioritizing CTS and ACK.

2. Focus Areas

- BSS vs. IBSS: definitions, minimum station count for IBSS, and the role of the DS in connecting multiple BSSs.
- Why 802.11 uses CSMA/CA instead of CSMA/CD: weak received signals, hidden terminal problem, and fading.
- CCA: power threshold (-82 dBm) and the two sensing variants; when the channel is declared busy vs. idle.
- NAV as virtual carrier sensing: how it implements channel reservation without physical sensing; when a station is prohibited from transmitting.
- Four IFS types: purpose of each (SIFS for ACK/CTS priority, PIFS for PCF, DIFS for DCF, EIFS for error recovery); their relative durations and the relationship $\text{PIFS} = \text{SIFS} + \text{aSlotTime}$ and $\text{DIFS} = \text{SIFS} + 2 \times \text{aSlotTime}$.
- Contention Window and BEB: initial CW, how CW grows after each failure ($\text{CW} = 2^{n-1} \cdot \text{CW}_{\min} - 1$), and the cap at $\text{CW}_{\max} = 1024$ for 802.11a.
- Backoff countdown rules: count down only while channel is idle after DIFS; freeze when busy; resume (do not restart) when idle again; transmit when counter reaches zero.
- RTS/CTS in 802.11: four-frame sequence ($\text{RTS} \rightarrow \text{CTS} \rightarrow \text{DATA} \rightarrow \text{ACK}$) all separated by SIFS; how the duration field in RTS and CTS sets NAV in neighbors and hidden stations.
- Hidden station solution using NAV: a hidden station may not hear RTS but will hear CTS and set its NAV accordingly.
- Medium access overhead sources: DIFS, backoff slots, RTS/CTS (optional), ACK, SIFS gaps, and frame headers; quantitative impact ($\approx 35\%$ overhead at 54 Mbps).
- DCF inefficiencies: idle channel during countdown, collision probability at high node density (exceeds 40% at 10 nodes choosing the same slot), and throughput degradation under congestion.
- EIFS: when it is used (after receiving a corrupted frame or when hidden-terminal scenarios lead to overlapping exchanges) and why it imposes a longer deferral than DIFS.

3. How to Study These Topics

1. Draw the 802.11 architecture diagram from memory: label the BSS, IBSS, DS, AP, portal, and the connection to an 802.X wired LAN.
2. Derive the IFS values for 802.11a from first principles: start with $aSlotTime = 9\mu s$ and $SIFS = 16\mu s$, then calculate PIFS and DIFS.
3. Draw the timing diagram for a successful DCF exchange from scratch: label every interval (DIFS, CW slots, SIFS, RTS, CTS, DATA, ACK).
4. Trace the BEB algorithm: given $CW_{min} = 16$ in 802.11a, compute the contention window after the 1st, 2nd, 3rd, and 4th failed attempts.
5. Simulate the backoff example from the lecture: AP1 with backoff = 15 and AP2 with backoff = 25; step through countdown, transmission, wait, and next round.
6. Draw the hidden-station RTS/CTS scenario with six stations as in the lecture; annotate which stations set NAV upon RTS and which set NAV upon CTS, and explain why station 6 defers.
7. List all sources of 802.11 medium access overhead and estimate qualitatively how each affects throughput as the number of active nodes increases.
8. Prepare one-page handwritten summaries of:
 - The complete DCF sender and receiver state machines.
 - The four IFS types with values, formulas, and use cases.
 - The BEB algorithm with the CW formula and a worked numerical example.

4. Exercises

4.1. Architecture and Standards

1. Draw and label the 802.11 architecture diagram. Define BSS, IBSS, DS, and portal. Explain how multiple BSSs form an Extended Service Set (ESS).
2. Compare DCF and PCF in terms of traffic type served, priority level, coordination mechanism, and whether they are mandatory or optional in 802.11.
3. What is the minimum number of stations required to form an IBSS? Justify your answer from the definition.
4. All 802.11 variants use CSMA/CA for multiple access. Explain why CSMA/CD, which is used in wired Ethernet, cannot be used in 802.11.

4.2. Clear Channel Assessment and Virtual Carrier Sensing

5. Define Clear Channel Assessment (CCA). What power threshold is used to declare the channel busy? State two variants of CCA described in the lecture.

6. Define the Network Allocation Vector (NAV). How does it differ from physical carrier sensing (CCA)? Under what conditions does a station refrain from initiating a transmission?
7. Explain why SIFS is too short for CCA. What function does SIFS serve at the hardware level (transceiver turnaround)?
8. A station checks both CCA and NAV before transmitting. Describe a scenario in which CCA reports the channel idle but NAV prevents the station from transmitting.

4.3. Timing and Interframe Spaces

9. List the four interframe spaces defined in 802.11 in order of increasing duration. State the purpose of each.
10. Using the 802.11a parameters ($aSlotTime = 9 \mu s$, $SIFS = 16 \mu s$), calculate PIFS and DIFS. Show your working.
11. Explain the priority mechanism encoded in IFS durations. Why does a shorter IFS give higher medium access priority?
12. Describe when EIFS is used instead of DIFS. What condition triggers it, and why does it impose a longer deferral?
13. Draw the timing diagram (from the lecture slides) showing a busy channel, deferred access, and the contention window with SIFS, PIFS, DIFS, and the RTS placement. Label each interval with its 802.11a value.

4.4. Frame Exchange Scenarios and ARQ

14. Describe the three MSDU delivery scenarios from the lecture (first unsuccessful, second unsuccessful, and successful). For each, state what happens at station 1 and station 2 and why.
15. In the second unsuccessful scenario, the DATA frame is received correctly by station 2, but the ACK is lost due to interference. What does station 1 infer? What action does it take?
16. Why does 802.11 use link-layer ARQ (ACK/retransmission)? What characteristic of wireless channels makes this necessary?
17. Explain why a station must wait SIFS before sending an ACK, while a new transmitter must wait DIFS before starting a fresh frame. What would happen if ACK used DIFS instead of SIFS?

4.5. Collision Avoidance and Binary Exponential Backoff

18. Describe the complete backoff procedure: when is it triggered, how is the random backoff count selected, and how does the countdown behave when the channel becomes busy?
19. Given $CW_{\min} = 16$ and $CW_{\max} = 1024$ (802.11a), compute the contention window after the 1st, 2nd, 3rd, 4th, and 5th failed transmission attempts.

20. Explain why a station that deferred during a busy period does not select a new backoff value but resumes its previous countdown. What fairness property does this provide?
21. Two stations, AP1 (backoff = 15) and AP2 (backoff = 25), both wait for the channel to become idle after DIFS. Trace the countdown step by step (RTS/CTS disabled). State which station transmits first and what the other station does while waiting.
22. Continue the above scenario: after AP1 finishes its Data+ACK exchange, AP1 draws a new backoff of 18 and AP2 resumes its remaining backoff of 10. Trace the next round to determine which station transmits next.
23. Why does BEB increase the contention window upon each failed attempt rather than keeping it fixed? What would happen to throughput if CW_{min} remained at CW_{min} regardless of collisions?

4.6. RTS/CTS and Hidden Station Problem

24. Explain how the RTS/CTS mechanism allows a transmitting station to “reserve” the channel. What information is carried in the duration field of an RTS or CTS frame?
25. Describe how the NAV is set in stations that are neighbors of the transmitter (heard RTS) and in stations that are hidden from the transmitter but close to the receiver (heard CTS only).
26. In the six-station example from the lecture, station 6 is hidden from station 2 but can hear station 1’s CTS. Explain step by step why station 6 does not transmit even though it never received the RTS.
27. Why is SIFS shorter than DIFS in the RTS/CTS exchange? What would happen if CTS used DIFS instead of SIFS after receiving an RTS?
28. State one advantage and one disadvantage of enabling RTS/CTS in 802.11. Under what conditions is the overhead of RTS/CTS justified?

4.7. Medium Access Overhead and DCF Inefficiencies

29. List all sources of overhead in a complete 802.11 DCF transmission with RTS/CTS enabled. Draw the transmitter and receiver timelines labelling each overhead component.
30. The lecture states that 802.11 DCF produces approximately 35% overhead at 54 Mbps. Explain which overhead components dominate at high data rates.
31. Explain the three significant inefficiencies of 802.11 DCF identified in the lecture: idle channel time during countdown, collision probability under high node density, and exponential backoff under congestion.
32. At 10 active nodes, the probability of at least two nodes choosing the same backoff slot exceeds 40%. Explain the consequences for throughput and latency.
33. Compare channel wastage caused by backoff (idle slots) and channel wastage caused by collisions (retransmissions). Which is worse under low node density? Which dominates under high node density?

5. Sample Questions

5.1. Short Questions

1. What does BSS stand for, and what is it in 802.11?
2. What is the role of the Coordination Function (CF) in 802.11?
3. State the difference between DCF and PCF in one sentence each.
4. What is the minimum number of stations in an IBSS?
5. State the frequency band and maximum data rate of 802.11n.
6. What is CCA, and what power threshold is used in 802.11?
7. Define the NAV. Is it a physical or virtual carrier sensing mechanism?
8. List the four IFS types in order from shortest to longest.
9. State the formula for DIFS in terms of SIFS and aSlotTime.
10. What are the 802.11a values of aSlotTime, SIFS, PIFS, and DIFS?
11. When is EIFS used instead of DIFS?
12. State the BEB formula for the contention window after the n -th failed attempt.
13. What are $CW_{\min}$ and $CW_{\max}$ in 802.11a?
14. Why does a station resume (not restart) its backoff countdown after the channel becomes idle again?
15. What information does the duration field in an RTS or CTS frame carry, and how do other stations use it?

5.2. Descriptive Questions

16. Describe the complete 802.11 DCF sender state machine. Include channel sensing, DIFS wait, backoff countdown, RTS/CTS exchange, data transmission, ACK wait, and the actions taken on success and failure at each step.
17. Explain the three MSDU delivery scenarios (first unsuccessful, second unsuccessful, and successful) using timing diagrams. For each scenario, state what frame is lost, what each station infers, and what action follows.
18. Explain the four interframe spaces (SIFS, PIFS, DIFS, EIFS) in detail. For each, state its purpose, its formula or value in 802.11a, and the scenario in which it is applied.
19. Describe the Binary Exponential Backoff (BEB) algorithm in detail. Explain the purpose of the contention window, how it grows after each failed attempt, how the countdown proceeds, and why deferred stations resume rather than restart their countdown.

20. Describe the RTS/CTS mechanism as specified in 802.11. Explain how consecutive frames are spaced, how the NAV is set in neighboring and hidden stations, and how SIFS prioritizes CTS and ACK frames.
21. Discuss the medium access overhead of 802.11 DCF. Identify all overhead components, explain their origins, and describe the three major inefficiencies of the DCF under high network load.

5.3. Analysis and Design Questions

22. A university deploys an 802.11a network. Station A has a backoff of 7 slots and station B has a backoff of 12 slots after the same busy period. Both start counting down simultaneously after DIFS. (a) Which station transmits first? (b) What is station B's remaining backoff when station A begins transmitting? (c) After station A's transmission ends, how many additional slots does B wait before it can transmit?
23. In an 802.11a network, $CW_{\min} = 16$. A station fails to transmit successfully on four consecutive attempts. (a) Compute the CW after each failure. (b) If the fifth attempt succeeds, what CW is used on the sixth attempt? Justify your answer.
24. Consider a network where RTS/CTS is disabled. Node A and node B are both hidden from each other but communicate with the same AP. Both sense the channel idle after DIFS and transmit simultaneously. (a) What happens at the AP? (b) How does the AP signal failure to A and B? (c) What does each node do next, and how does BEB reduce future collision probability?
25. Consider the same hidden-terminal scenario but with RTS/CTS enabled. Node A sends an RTS to the AP first (winning the backoff). (a) Trace every frame in the exchange and identify which nodes set NAV upon each frame. (b) Show why node B cannot cause a collision during A's DATA transmission.
26. A dense office network has 20 simultaneously active 802.11g stations all within range of one AP. Explain why DCF performs poorly in this scenario. Discuss the contribution of BEB to the problem and propose one protocol-level improvement that could increase throughput.
27. Evaluate the trade-off between enabling and disabling RTS/CTS in an 802.11 network. Construct a table comparing the two modes across the following criteria: hidden terminal protection, overhead per packet, effective throughput for small packets, effective throughput for large packets, and NAV-based protection. Under what traffic profile does each mode perform better?